Morteza Sabri

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Portfolio | LinkedIn | GitHub | Google Scholar | ORCID

PROFESSIONAL SUMMARY

Clinical Bioinformatician (M.Sc. Biology, Genetics) with expertise in cancer genomics, NGS pipeline engineering, and Molecular Tumor Board (MTB) infrastructure. Currently at Universitätsmedizin Greifswald, enhancing and maintaining **MIRACUM-Pipe** (somatic variant pipeline for whole-exome and targeted panel sequencing) in collaboration with University Medical Center Rostock, automating TSO500 somatic panel workflows on Illumina ICA/ICI, and developing methylome-based glioblastoma classification pipelines. Extensive R programming experience since 2017 with proven expertise in bulk and single-cell transcriptomics, methylation arrays, and reproducible pipeline development. Author of **goiExplorer**, an R package for interactive gene-of-interest exploration.

HIGHLIGHTED AREAS OF EXPERTISE

- Clinical cancer genomics & computational oncology
- Bulk & single-cell RNA-Seq
- Methylome analysis & epigenomics
- MTB infrastructure & somatic variant analysis
- R programming & shell scripting
- Workflow automation (Nextflow, Docker, ICA CLI)
- Data visualization & reproducibility
- Study design & data interpretation

EXPERIENCE

Scientific Staff, Bioinformatician | Universitätsmedizin Greifswald, Greifswald *Oct 2025 – Present*

- Enhancement and maintenance of MIRACUM-Pipe (somatic variant pipeline for whole-exome and targeted panel sequencing: SNV/InDel/CNV/LoH) in collaboration with University Medical Center Rostock for clinical MTB use
- Automation of TSO500 somatic panel workflows on Illumina ICA/ICI (sample sheet validation, DRAGEN pipeline, QC filtering, audit tracking)
- Methylome-based glioblastoma subgroup classification using EPIC arrays and Heidelberg CNS classifier
- AI-assisted oncogenic evidence aggregation for Molecular Tumor Board reporting
- Institutes: Institute of Molecular Genomics & Institute of Bioinformatics

Scientific Staff, Bioinformatician | RWTH Aachen University, Aachen *Jan 2025 – Apr 2025*

- Performed single-cell RNA-Seq analysis to study neuron migration in the absence of DNMT1
- Conducted methylation array data analysis on 64 subjects with postpartum depression
- Handled data preprocessing (Parse Bioscience's Trailmaker), statistical analysis, and visualization

Scientific Staff, Bioinformatician | Technical University of Munich, Munich *Nov 2023 – Nov 2024*

- Conducted bulk and single-cell RNA-Seq analyses for cardiovascular research
- Managed workflows from raw data to functional interpretation and identifying new biomarkers
- Generated publication-ready visualizations illustrating gene-disease associations
- Optimized preprocessing pipelines with R/Shell scripting, reducing runtime up to 80%
- Performed deconvolution analysis using BayesPrism R package to elucidate cell-type composition

Visiting Scientist | Helmholtz Munich, Munich *Oct 2024*

- Conducted deconvolution analysis on scRNA-Seq data

Instructor & Freelance Bioinformatician | Self-Employed, Tehran *2019 – 2023*

- Delivered over 10 onsite workshops on R programming, RNA-Seq, and Linux/Shell scripting
- Provided freelance bioinformatics services including RNA-Seq, epigenomics, and visualization
- Supported manuscript revisions by performing advanced statistical analyses (e.g., MINT)
- Hosted *Biocast101* podcast (Apple, Spotify, Castbox, PodLink)
- Created content on social media as *Bioinformatics101* (YouTube, Instagram, etc.)

Compulsory Military Service | Department of Defense, Tehran *2020 – 2022*

- Sampled blood and oral swabs (COVID-19), performing CBC diagnostics
- Managed patient reception, communication, and scheduling

Research Assistant | National Institute of Genetic Engineering and Biotechnology (NIGEB), Tehran *2019 – 2020*

- Produced publication-quality visualizations (heatmaps, volcano plots) to support manuscript figures
- Co-authored a peer-reviewed article on monolignol biosynthesis in *Camelina sativa*
- Conducted bulk RNA-Seq analyses on *Camelina sativa*

Graduate Researcher | University of Sistan and Baluchestan, Zahedan *2017 – 2019*

- Designed and executed qRT-PCR experiments comparing drought responses in two grapevine cultivars
- Performed statistical analyses and produced publication-quality visualizations
- Authored grant proposals on drought-stress gene-expression profiling and medicinal plant genomics
- Lead-authored Master's thesis on drought-stress gene expression, overseeing study design and data interpretation
- Supervised two cohorts of B.Sc. students on *Drosophila melanogaster* genetics projects
- Analyzed publicly available bulk RNA-Seq and miRNA-Seq datasets to identify stress-responsive genes
- Co-authored a peer-reviewed article on trastuzumab resistance in HER2-positive breast cancer

SKILLS

Lab Techniques	RNA/DNA isolation, PCR/qRT-PCR, primer design and optimization, electrophoresis
Programming	R (proficient), Python (basic), MATLAB (basic), Shell/Bash, Git/GitHub, LaTeX, Markdown
Infrastructure	Nextflow, Docker/Singularity, Illumina ICA/ICI CLI, workflow automation
Bioinformatics	Bulk RNA-Seq, scRNA-Seq, methylation arrays, ChIP-Seq, Methyl-Seq, TSO500 somatic panel, WES (MIRACUM-Pipe), somatic variant calling & annotation
Software	CLC Genomics Workbench, Galaxy, SPSS, JupyterLab, Trailmaker, VS Code
HPC Platforms	Sherlock cluster (Stanford University, via collaboration), RWTH Aachen cluster, Greifswald HPC
Transferable Skills	Scientific writing, teaching & mentoring, workshop design & delivery, open science & reproducibility, FAIR data management, teamwork & communication

EDUCATION

M.Sc. Biology (Genetics) | University of Sistan and Baluchestan, Iran *2015 – 2019*

- Thesis: "Investigation of Superoxide Dismutase and Catalase Gene Expression under Drought Stress: A Comparative Study between Sistan and Baluchestan and Moderate Cultivars"

B.Sc. Cellular and Molecular Biology (Genetics) | Islamic Azad University, Iran *2011 – 2014*

CERTIFICATIONS & LANGUAGES

Certifications: Programming Foundations (Fundamentals, Beyond the Fundamentals, Fuzzy Logic), Learning Bash Scripting — LinkedIn Learning (certificates)

Languages: Persian (Native), English (Proficient), Azeri (Basic)

PUBLICATIONS

- Overcoming trastuzumab resistance in HER2-positive breast cancer. *Journal of Cellular Physiology*, 2019. DOI: 10.1002/jcp.29216
- Evaluation of monolignol biosynthesis gene network in *Camelina sativa*. *Agricultural Biotechnology Journal*, 2020.
- Investigation of Superoxide Dismutase and Catalase Gene Expression under Drought Stress: A Comparative Study between Sistan and Baluchestan and Moderate Cultivars. *(Manuscript in preparation)*

TEACHING EXPERIENCE

Workshops: R & Linux/Shell programming, RNA-Seq analysis, Visualization and basic statistics in R, Introduction to R for Biologists.
See portfolio for details.

HIGHLIGHTED PIPELINES

- Somatic Variant Analysis (WES & targeted panel) — MIRACUM-Pipe (BWA-MEM → Mutect2/VarScan2/CNVkit → ANNOVAR/CIViC → MTB PDF report)
- TSO500 Somatic Panel — Illumina ICA/ICI automation (Sample sheet validation → DRAGEN → QC → clinical variant report)
- Methylome Classification — Glioblastoma subgroups (IDAT → minfi/ChAMP → Heidelberg CNS classifier → subgroup report)
- Bulk RNA-Seq / polyA / ncRNA-Seq (STAR/HISAT2/Salmon → DESeq2/edgeR/limma → GO/KEGG/ReactomePA)
- scRNA-Seq — cell and nuclei (10x Genomics / Parse Biosciences → Seurat/SingleCellExperiment → clustering/DE/deconvolution)
- Epigenomic arrays (IDAT → minfi/ChAMP → DM analysis → GO/KEGG)
- Methyl-Seq & ChIP-Seq (FASTQ → STAR → peak calling → ChIPseeker/DESeq2 → GO)
- Genome assembly (Illumina FASTQ → Velvet/SPAdes → QUAST scaffold assessment)

See portfolio for details.

SOFTWARE DEVELOPMENTS

MIRACUM-Pipe (maintenance & clinical adaptation) — github.com/AG-Boerries/MIRACUM-Pipe

- Somatic variant pipeline for whole-exome and targeted panel sequencing, originally developed by AG Boerries (University Medical Center Freiburg); maintaining and adapting for the MTBs and CCCs at Universitätsmedizin Greifswald and University Medical Center Rostock (QC tuning, reporting modules, institutional workflow integration)

goiExplorer (R package) — github.com/mortezasabri/goiExplorer

- R package for interactive gene-of-interest exploration from count matrices or Salmon (quant.sf) outputs, with built-in plotting functions, full documentation, and unit tests